

EVENTIFY

Event Hall Booking & Management System

A National Diploma Project

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Problem Statement

Why Eventify is needed

01 Manual Booking Process

Event halls in Nigeria rely on phone calls and physical visits, leading to double bookings and scheduling conflicts.

02 No Real-Time Availability

Customers cannot check hall availability instantly, causing delays and frustration.

03 Payment Tracking Issues

Cash-based payments lack transparency, leading to disputes and poor record keeping.

04 No Centralized Management

Admin tasks like managing catering, decorations, and reminders are handled separately with no unified platform.

Project Objectives

What Eventify aims to achieve

- 1 Design and implement a web-based event hall booking system with real-time availability checking.
- 2 Integrate a secure online payment gateway (Paystack) for seamless transaction processing.
- 3 Develop an AI-powered package recommendation engine to assist users in event planning.
- 4 Build an administrative dashboard for managing bookings, halls, catering, and decorations.
- 5 Implement automated email notification and reminder systems for upcoming events.

Literature Review

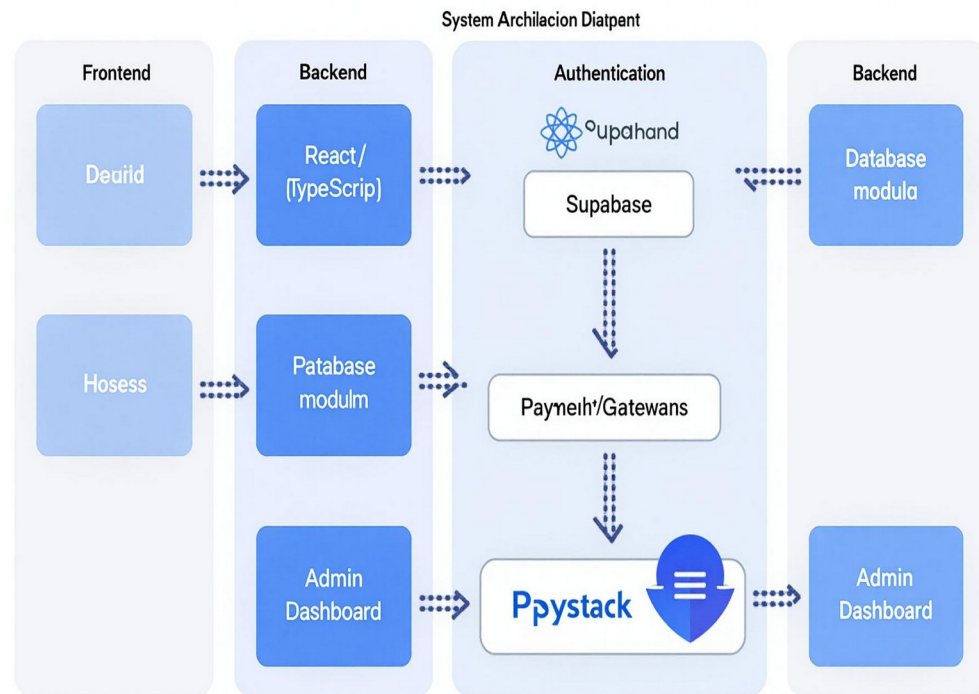
Comparison with existing solutions

Feature	Traditional Methods	Generic Booking Platforms	Eventify
Real-time Availability	✗	✓	✓
Online Payment	✗	✓	✓ (Paystack)
AI Recommendations	✗	✗	✓
Catering & Decorations	✗	✗	✓
Automated Reminders	✗	Limited	✓
Budget Calculator	✗	✗	✓
Role-Based Admin	✗	Limited	✓

Eventify fills a critical gap by combining hall booking with catering, decorations, AI recommendations, and integrated payments — features absent in existing Nigerian solutions.

System Architecture

Technology stack and infrastructure



React + TypeScript

Frontend Framework

Tailwind CSS

Styling & Design System

Supabase

Backend & Database

Paystack

Payment Gateway

Edge Functions

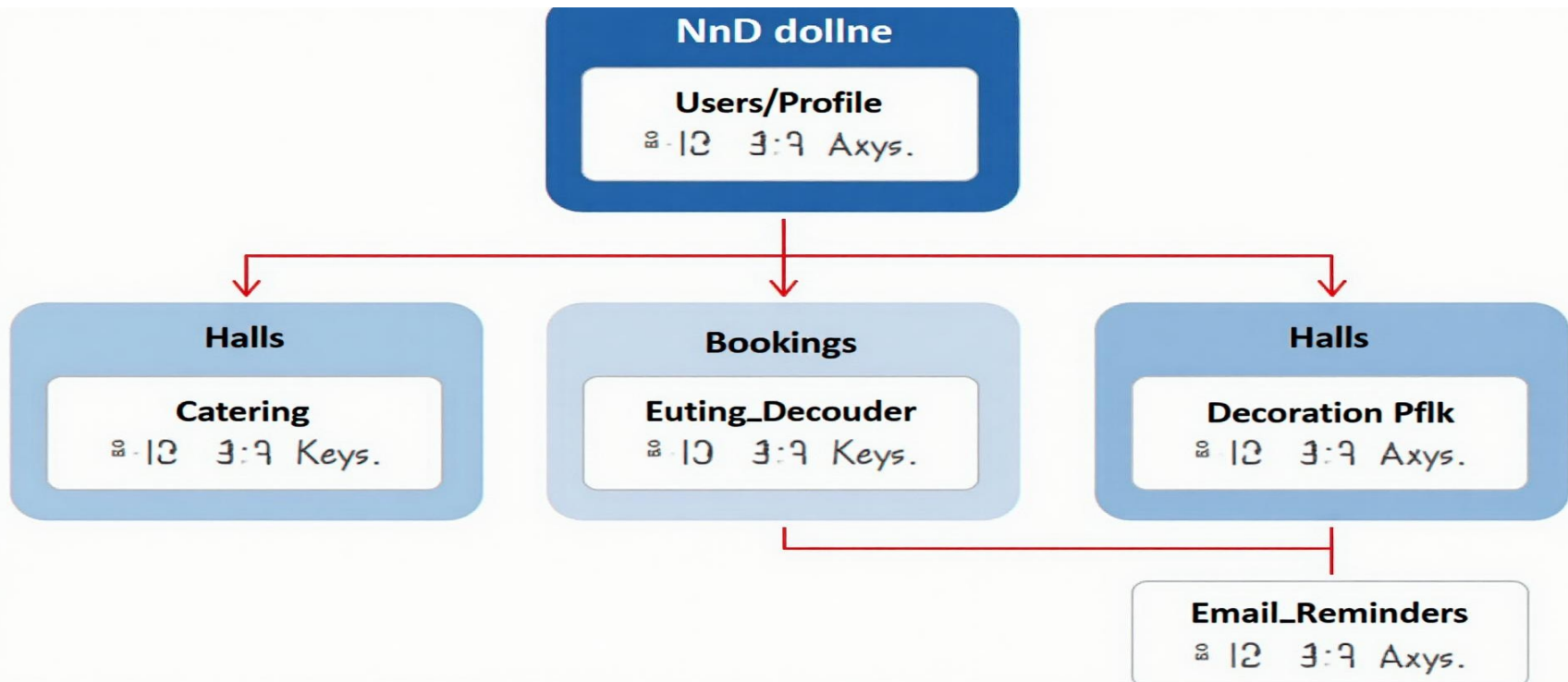
Serverless Backend Logic

Tanstack Query

Data Fetching & Caching

Entity Relationship Diagram

Database schema design



11 interconnected tables including bookings, halls, catering_packages, decoration_packages, profiles, notifications, reviews, user_roles, and email_tracking.

Key Features (1/3)

Core booking and planning features

Hall Booking with Real-Time Availability

- Browse available event halls with capacity and amenities
- Date and time-based availability checking to prevent conflicts
- Dynamic pricing based on hours and hall selection

Catering & Decoration Packages

- Pre-configured catering packages with per-person pricing
- Decoration packages with style categories and feature lists
- Seamless selection during the booking flow

Smart Budget Calculator

- Real-time cost estimation as users select options
- Breakdown by hall, catering, and decoration costs
- Helps users plan within their budget constraints

Key Features (2/3)

Payment, AI, and automation

Paystack Payment Integration

- Secure online payments via Paystack gateway
- Payment verification through Edge Functions
- Automatic status updates (pending → paid → refunded)
- Tiered refund policy based on cancellation timing

AI Package Recommender

- Powered by Gemini AI model
- Analyzes event type, guest count, and budget
- Suggests optimal hall, catering, and decoration combinations
- Personalized recommendations via Edge Function

Automated Email Reminders

- Scheduled reminders at 7-day, 3-day, and 1-day intervals
- Booking confirmation notifications
- Email tracking with open and click analytics
- Cron-triggered Edge Function execution

Key Features (3/3)

Administration and user experience

Admin Dashboard

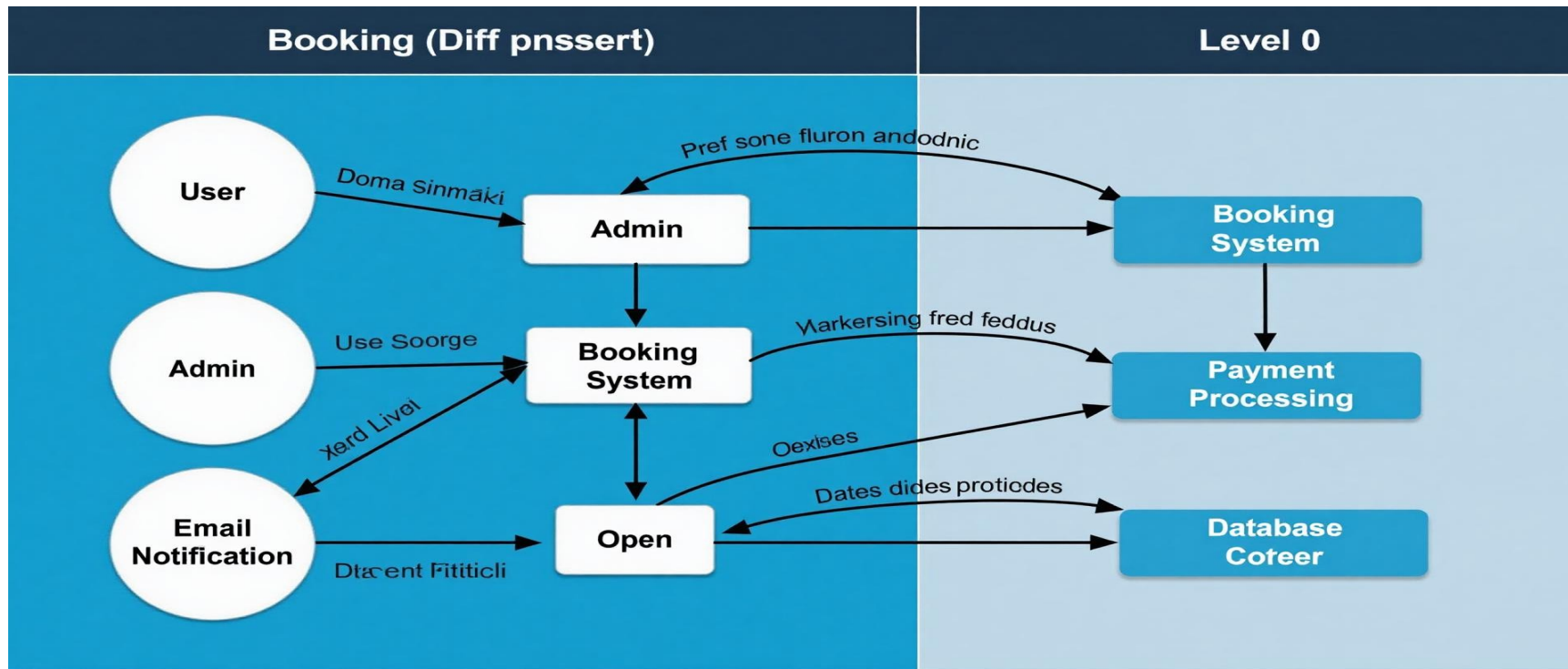
- Manage halls, catering, and decoration packages
- View and manage all bookings with status filters
- Review moderation system (approve/reject)
- User role management (admin, moderator, user)
- Email delivery analytics and tracking

User Dashboard

- View personal booking history and details
- Track payment status and event dates
- Reschedule bookings with availability check
- Leave reviews for completed events
- Real-time notification bell with unread count

Data Flow Diagram

How data moves through the system



Security & Access Control

Protecting data and ensuring proper authorization

Row-Level Security (RLS)

- Every database table protected with RLS policies
- Users can only access their own bookings, profiles, and reviews
- Public data (halls, packages) accessible without authentication

Role-Based Access Control

- Separate `user_roles` table (not on profiles) prevents privilege escalation
- Security-definer `has_role()` function prevents recursive RLS
- Three roles: admin, moderator, user with distinct permissions

API & Edge Function Security

- JWT verification on sensitive endpoints
- Service role key validation for cron jobs
- CORS headers configured for cross-origin requests
- Input validation with Zod schemas on all forms

Testing & Results

Functional testing outcomes

Test Case	Description	Result
User Registration & Login	Email/password auth with verification	Passed
Hall Browsing & Selection	Filter by capacity, view amenities	Passed
Booking Creation	Date, time, hall, catering, decorations	Passed
Payment Processing	Paystack initialize & verify flow	Passed
AI Recommendations	Package suggestions via Edge Function	Passed
Admin Management	CRUD on halls, packages, bookings	Passed
Email Reminders	Automated 7/3/1 day notifications	Passed
Review System	Submit, approve, display reviews	Passed
Responsive Design	Mobile, tablet, desktop layouts	Passed

Challenges & Future Work

Limitations encountered and recommendations

Challenges

- Email deliverability in production environments
- Complex RLS policy debugging and recursive query prevention
- Payment webhook reliability and idempotency handling
- Managing real-time state synchronization across users

Future Work

- Mobile app development (React Native)
- SMS notifications via Twilio integration
- Virtual hall tours with 360° images
- Multi-vendor marketplace for service providers
- Advanced analytics and reporting dashboard

Conclusion

Eventify successfully demonstrates a full-stack web application that addresses the challenges of manual event hall booking in Nigeria. By combining real-time availability, secure payments, AI-powered recommendations, and comprehensive admin tools, the system provides a modern, scalable solution for event management.

Thank You

Questions & Answers

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